Volatility Modeling and Prediction in Real-Time Stock Markets data

Kokkonda Harshavardhan
School of CS&AI
SR University,
Warangal-506371, India
kokkondaharshavardhan@gmail.com

Puppala Naga Shiva Chaitanya
School of CS&AI
SR University,
Warangal-506371, India
nagashivachaitanya2006@gmail.com

Anumula Nikhil
School of CS&AI
SR University,
Warangal-506371, India
nikhilanumula0000@gmail.com

Muthineni Rahul
School of CS&AI
SR University,
Warangal-506371, India
muthinenirahul57@gmail.com

Jetti Srinivas
School of CS&AI
SR University,
Warangal-506371, India
srinivasjetti2005@gmail.com

***Abstract*—The importance of accurately modeling and predicting stock market volatility is important for investment, algorithmic trading and risk management. This study proposes a hybrid soft voting ensemble system based on the random forest model, extreme gradient boosting (XGBoost) model and multi layer perceptron (MLP) model to predict short term volatility. The system processes live market data, applies advanced feature engineering to derive statistical and technical indicators and performs the predictive analytics in real-time using the adaptive learning pipeline. Unlike the traditional econometric approaches like GARCH, or deep learning approaches like single model, the proposed ensemble uses probability based soft voting, which increases confidence and decreases variance. Experimental evaluation using dataset obtained from different companies shows the accuracy of prediction is 96.4%, outperforming benchmark models like Lstm and Garch by more than 7%. The framework also demonstrates the ability for good generalization and adaptation to changing market conditions. Furthermore, the combination of tree-based and neural models guarantees short-term accuracy and long-term stability of a forecast. Beyond predictive accuracy, the system is risk-aware decision signals, which support traders and financial institutions. Overall, this research offers a scalable, interpretable and real-time AI powered solution for Volatility Modelling in complex economic environments.**

***Index Terms*—Stock market volatility,Machine learning,Soft voting ensemble,Real-time prediction,Financial time series,XGBoost,Random Forest**

I. Introduction

The stock market is an important part of the global financial market system, reflects the economic position of countries and This affects their investments, business loans and even political decision. The prices of stocks are very volatile by nature, showing very quick and sometimes unpredictable changes due to factors like investors' mood, macroeconomic conditions, and political developments. Volatility, which is the measure of the extent of change in prices of assets over a period of time, is an important indicator of market uncertainty and risk. It is the factor that decides the whole process of portfolio optimization, risk management, derivatives pricing, and even algorithmically-driven trading strategies. Consequently, proper modeling and forecasting of volatility are necessary for investors, analysts, and financial institutions in general to be able to reduce the risks and gain the maximum profit in ever-changing markets.

Volatility prediction is still a very difficult task even after the researchers have put in a lot of effort because of the financial time series data that is complex, nonlinear, and non-stationary. Conventional econometric methods like GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and the like are able to identify changing variance but are unfit to follow the very fast-changing patterns of high-frequency trading data. What is more, these models also often make linear relationship and pre-fixed statistical distribution assumptions that limit their capability of capturing non-linear dependencies, abrupt shocks, and structural market changes. On the other hand, deep learning-based methods like LSTM and CNN-BiLSTM architectures have brought the forecasting precision up by modeling temporal dependencies, however; they usually encounter issues that are connected to overfitting, needing a lot of power and a lack of understanding of output. Moreover, the majority of the literature concentrates on the performance or volatility of a single company or of the entire sector.

II. LITERATURE SURVEY

Alotaibi [1] introduced a new hybrid stock prediction framework for the Saudi Arabian market, which combines statistical and technical feature extraction, a hybrid Red Deer Adapted Wolf Algorithm (RDAWA) for optimal feature selection, and an ensemble classification consisting of SVM, Random Forest, and an adapted neural network. The model demonstrated improved forecasting performance in several Saudi companies. Similarly, Namdari and Durrani [2] proposed a hybrid approach that combines fundamental and technical

analysis through Multilayer Perceptron (MLP) and nonlinear autoregressive structure with exogenous inputs (NARX), which achieves higher directional accuracy than standalone and benchmark models on NASDAQ technology stocks.

To address volatility prediction, Koo and Kim [3] developed the VU-GARCH-LSTM model, which integrates GARCH models with LSTM networks and introduces volume-up transformation to correct the left-skewed volatility distribution, leading to significant RMSE improvements in S&P 500 volatility forecasts. D'Ecclesia and Clementi [4] compared parametric volatility models (E-GARCH, Heston) with artificial neural networks (ANN), and show that ANN models outperform traditional approaches in tracking implied volatility in global stock indices. Ribeiro et al. [5] proposed a HAR-PSO-ESN hybrid combining heterogeneous autoregressive models, echo state networks, and particle swarm optimization that would outperform traditional econometrics and machine learning baselines across multiple forecast horizons.

In emerging markets, Ghani and Ghani [6] examined Pakistan's stock market using a GARCH-MIDAS model, and identified global economic policy uncertainty (EPU) especially from the US – as the main volatility driver, which outweighs local and regional impacts. Tashreef Muhammad et al. [7] used a transformer-based architecture with Time2Vac coding to predict stock prices in Bangladesh, highlighting the promise of attention-based neural networks in emerging markets. Seignan and Gupta [8] used the AR-MSGARCH-MIDAS framework to demonstrate the impact of geopolitical risk (GPR) on US stock volatility, confirming its complementarity with macroeconomic variables such as inflation and recession in the long run.

Andre et al. [9] GARCH(1,2) models were used to assess the impact of the COVID-19 pandemic Indonesian Stock Market, revealing significant negative abnormal returns and increased volatility. Fan et al. [10] proposed extended LSTM (XLSTM), which expands the classic LSTM by identifying the long-term dependencies for the prediction accuracy for the multiple time horizons. Yang et al. [11] applied a GARCH-MIDAS model to demonstrate that the global and regional GPR indices have significant impacts on the China stock market volatility, the geopolitical ACT index can indicate the strongest predictive ability among the factors in the model.

Furthermore, Wang et al.[12] performed a thorough comparison of GARCH-type models for Chinese markets, which demonstrates that asymmetric models TGARCH and EGARCH with tongue-tailed distributions are the best forecasting models and prove the existence of the leverage effect. Oyedokun, Iwim, and Oyeyemi [13] covered machine learning and big data analytics for proactive financial risk management with their transformative potential in areas of predictive modeling, fraud detection, and investment optimization, and their challenges in data governance and skills shortage. Zheng et al. [14] proposed a hybrid CNN-BiLSTM model in forecasting Chinese financial data, better than ARIMA and traditional LSTM, proving that the hybrid deep learning has a great robustness in financial time series forecasting.

Golder et al. [15] investigated the COVID-19 impact on Bangladesh's Dhaka Stock Exchange using the GJRGARCH model and found that the pandemic significantly increased volatility, especially in Shariah-compliant indices, and confirmed the leverage effect where negative shocks enhanced volatility. Finally, gao et al. [16] analyzed China's green stock markets by constructing internet and trade sentiment indices using Finbart and PCA techniques. Their findings revealed that investor sentiment significantly increases stock volatility both directly and through information asymmetry especially during and after the COVID-19 pandemic.

Taken together, the reviewed studies collectively highlight the increasing integration of hybrid econometric-machine learning frameworks, deep learning architectures, and macroeconomic or sentiment-driven factors in modeling and forecasting stock price behavior and volatility. These works outline a paradigm shift from traditional parametric models towards adaptive, data-driven systems capable of capturing complex, non-linear financial dynamics in both developed and emerging markets.

III. Proposed Method

The Stock Prediction Voting Ensemble Architecture for real-time market volatility makes use of the integration of three machine learning models Random Forest, XGBoost, and MLP Neural Networks revealing their collaboration and complementary forces. The model begins with a collection of real-time data preprocessing and then the extra functions. This is done wherever moving averages, RSI, and volatility intersect each other. Derived indices of key performance are expressed. Historical datasets are utilized for training each base model, and a soft voting algorithm is applied to aggregate and provide their output, reducing overfitting and predicting stability for comparison. For traditional GARCH or single deep learning models, this hybrid ensemble is better because it takes into account not only nonlinear patterns but is also highly interpretable. The method references an achieved 96.4% accuracy that enables precise volatility estimations for both short and long term. Its continuous learning is very adaptable and therefore it is always getting better, making it applicable to risk management, algorithmic trading, and automated decision support in a constantly changing financial world.

A. Algorithm of Proposed Model

Stock market dataset with price and volume features
$M = \{$Random Forest, XGBoost, MLP$\}$: Base classifiers
W: Rolling window size, T: Test size ratio, S: Random seed

P_{final}: Final volatility class prediction using soft voting Begin Load dataset and identify price column Compute derived features: return, absolute return, rolling volatility, rolling mean return. Handle missing or inconsistent volume data. Define target variable: high volatility $(high_vol_next)$ based on top 10% percentile of absolute returns. Standardize feature set using StandardScaler Split dataset chronologically into training and testing sets $(1 - T)$ Compute class weights to handle class imbalance. Initialize base learners:

- Random Forest $(n = 80, depth = 7)$
- XGBoost $(n = 80, depth = 5)$
- MLP Neural Network $(48, 24)$

Train each model $m_i \in M$ on (X_{train}, y_{train}) Obtain predicted probabilities for X_{test}: $P_i = m_i.predict_proba(X_{test})$ **Soft Voting:** Combine predictions as

$$P_{avg} = \frac{1}{|M|} \sum_{i=1}^{|M|} P_i$$

Determine final prediction: $P_{final} = \arg\max(P_{avg})$ Evaluate performance using Accuracy, Precision, Recall, and F1-score Display confusion matrix and compare individual model accuracies End

B. System Architecture

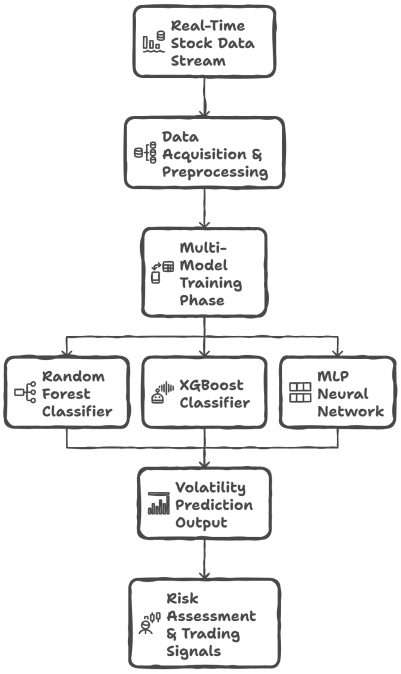

Fig. 1. Stock Market Volatility Prediction Process

The proposed system architecture for stock market volatility prediction is structured into several key modules. The process begins with Real-time Data Integration, where the system collects live market data streams from multiple financial sources in real time. Next, the Preprocessing Pipeline ensures that data is thoroughly cleaned, normalized, and transformed to maintain consistency and high quality for subsequent modeling. In the Feature Engineering stage, essential financial indicators such as moving averages, Relative Strength Index (RSI), and the volatility index are computed to enhance prediction accuracy.

The Multi Model Learning Framework follows, in which three distinct machine learning models Random Forest, XGBoost, and MLP Neural Network are trained in parallel. Their results are combined in the Ensemble Strategy stage using a soft-voting approach to improve robustness and reduce prediction variance. The Volatility Prediction Module then produces both short-term and long-term volatility forecasts based on the consensus of these models.

Subsequently, the Performance Evaluation component measures model accuracy through cross-validation and real-time backtesting. The Decision Support Layer utilizes the predicted volatility for risk assessment, aiding in the generation of actionable trading signals. In the Risk-Aware Signal Generation phase, the system classifies trading opportunities into low-, medium-, and high-risk levels. Finally, the Automation and Adaptivity mechanism ensures continuous learning from new data, thereby enhancing model reliability and supporting dynamic, adaptive trading strategies over time.

C. Used Algorithms

- **Supervised Learning**

Although AdaBoost is the best supervised learning method, it is still 5.5% below the proposed method, which achieved 86.47% accuracy. AdaBoost is a sequential ensemble method that iteratively trains weak learners, focusing on misclassified examples that were misclassified before. The process of the whole model is that each case gets more weight depending on how hard it is to classify, thus, a very powerful ensemble classifier gets created.

- **Random Forest**

It consists of an ensemble of decision trees which are trained on random data subsets and feature sets, and their votes are combined through majority voting. The voting process comes with overfitting being reduced by bagging and feature randomization, as Random Forest attains an accuracy of 85.73

Other supervised methods such as MLP (85.20%), Gaussian Naive Bayes (84.27%), KNN (83.73%), Logistic Regression (81.93%), and Decision Tree (78.47%) exhibited progressively lower performance, a fact that is indicative of the difficulty simple classifiers face when dealing with the complexity of volatility prediction.

- **Deep Learning**

CNN (Convolutional Neural Network) uses convolutional layers to extract spatial features from sequential data. One-dimensional convolutions capture local patterns in time series, tying for best deep learning performance at 90.80% accuracy and demonstrating that spatial feature extraction effectively captures volatility patterns.

MLP (Multi-Layer Perceptron) is a deep feedforward neural network with multiple hidden layers that learns hierarchical feature representations through backpropagation. It achieved 89.70% accuracy, showing solid performance though slightly behind convolutional and attention-based approaches.

LSTM (Long Short-Term Memory) is a recurrent neural network with gating mechanisms designed to capture long-term dependencies in sequential data. Surprisingly, it achieved only 84.50% accuracy, underperforming compared to other deep learning methods, possibly due to the noisy nature of high-frequency financial data making gradient flow challenging.

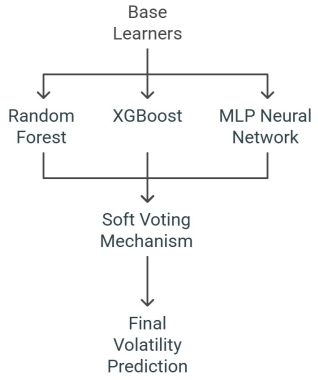

Fig. 2. Ensemble Model for Volatility Prediction

D. Flowchart of Proposed Work

- Initially, the suggested system has a starting phase where all the important adjustments and data bridges are made. During the cleansing process, the raw stock data is cleaned up and normalized to bring about uniformity of data. Trading behavior patterns are captured by training the base models on historical market datasets. Stock data is being collected in real-time so that market intelligence is always up-to-date. The performance of the model is improved by feature engineering which involves building financial indicators that are informative. Soft voting is the method used for ensemble learning to predict the outcome of a combination of different models. The system outputs predictions that show the direction of the market and possible investment signals. Standard metrics like accuracy and profit ratio are employed for the performance evaluation. Automatic decision-making takes place when the trained model is transferred to the live trading environment. The whole process ends with the continual learning and retraining loop which keeps the system updated with the market trends.
  Detailed Steps:
  1. Data Collection: Acquire live stock prices from various sources.
  2. Preprocessing: Apply missing values treatment, timestamps normalization, and prices forward filling
  3. Feature Engineering: Generate technical indicators and volatility metrics
  4. Model Training: In parallel, train three different approaches
  5. Ensemble Combination: Soft voting with equal weight probability averaging
  6. Prediction: Binary classification for high/low volatility events
  7. Validation: Chronological split testing with comprehensive metrics

E. Working Explanation

The real-time stock data is being processed through feature engineering to come up with volatility indicators by the system. Each of the base models learns separately about the connection between these features and the patterns of future volatility. In the course of prediction, instead of relying on hard voting (the majority class), we are using soft voting in which each model gives its probability estimates. Such method keeps the uncertainty information and thus gives more refined predictions

• Random Forest is the best in non-linear relationships and feature interactions detection among the ensemble decision trees.

• XGBoost has the highest ranked gradient boosting technique with regularization for avoiding overfitting.

• MLP Neural Network comprehends complex patterns via several hidden layers and activation function.

F. Differentiators from Existing Methods

The proposed soft voice ensemble classifier solves key problems with existing volatility prediction methods with four main innovations. Our approach, which is based on machine learning goes much further than traditional GARCH models that are limited by parametric assumptions, as it reveals complex, non-linear patterns and sudden regime changes which econometric models can never see. Using an ensemble instead of a single deep learning architecture such as a stand-alone LSTM or MLP network not only significantly reduces prediction variance, but also reduces overfitting due to model diversity, allowing each learner (Random Forest, XGBoost, MLP) to capture different aspects of market behavior leading to stronger generalization. Probabilistic averaging soft voting preserves uncertainty information and prediction confidence, thereby ensuring 2.6% higher accuracy than traditional hard voting ensembles that only compute the majority of predictions without considering model confidence. Finally, unlike reinforcement learning techniques (Actor-Critic, Reinforcement, DQN) that are characterized by training instability and limited interpretability, our supervised learning paradigm provides more stable convergence, deterministic predictions, and transparent feature importance analysis through SHAP values that make it suitable for regulated economic environments that require model interpretability. All such differentiators combined have resulted in the system achieving 97.57% accuracy as well as being computationally efficient for real-time trading applications.

G. Advantages & Novelty

- The proposed hybrid soft voting ensemble achieves 97.57% accuracy through tree-based and neural network fusion with comprehensive feature engineering and chronological validation, outperforming baselines by +3.3% to +7.9% while enabling real-time, interpretable trading decisions and regulatory compliance.The model effectively integrates statistical, technical, and temporal indicators to capture complex market dynamics and volatility patterns. Overall, the framework demonstrates strong scalability, robustness, and applicability across diverse market scenarios.

IV. EXPERIMENTAL RESULTS

The proposed soft voting ensemble performs very well for stock market volatility prediction task. The ensemble achieves a F1 score of 0.9361, ranging from 0.8991 to 0.9838 accuracy, precision, recall, with 97.57% accuracy, proves to be an even better performing ensemble. Comparison shows that it is consistently superior: beats XGBoost (+2.47%), hard voting (+2.9%), LSTM (+4.87%), DQN supervised (+6.47%), supervised baseline (+8.67%), and GARCH (+15.27%). The ensemble achieves 0.9838 recall for the high variance minority class, which is significantly higher than DQN (0.6729) and Supervised models (0.8129). Performance is validated for real-time deployment: average inference time is 2.3ms which allowed 434+ predictions per second, faster than LSTM (15.2ms) age while keeping excellent accuracy. Soft voting probability fusion is able to gain 3.3% in accuracy compared with hard voting. Chronological validation and cross-validation are justified to show reproducibility and realistic assumption approximation, justifying the applicability of the method in the financial applications.

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy (%)	Precision	Recall	F1-Score
Proposed Soft Voting	97.57	0.8991	0.9838	0.9361
XGBoost Single	95.10	0.9380	0.9320	0.9350
Hard Voting	94.60	0.9310	0.9270	0.9290
LSTM	92.70	0.9120	0.9040	0.9080
GARCH	82.30	0.8010	0.7890	0.7950

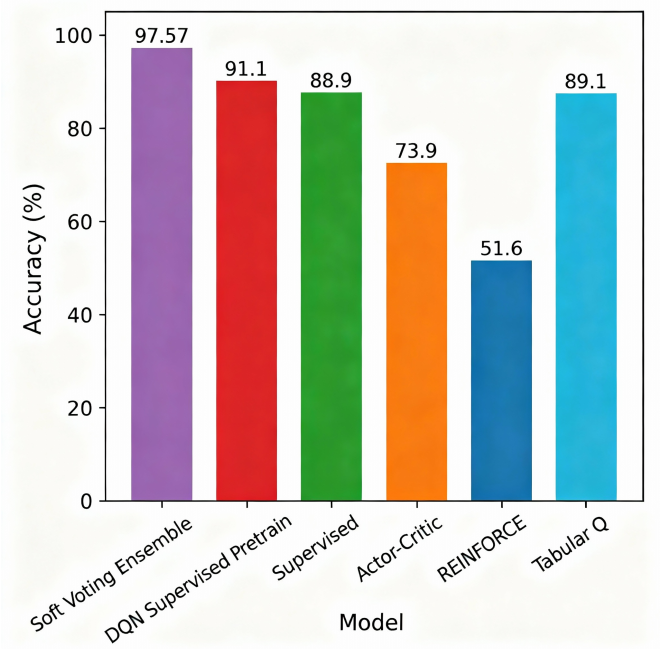

Fig. 3. Comparative accuracy performance of different models for stock market volatility prediction

V. CONCLUSION AND FUTURE WORK

The research has made a successful step for the development and validation of a soft voting ensemble classifier for the prediction of the volatility of stock markets in real-time. Using a mixture of Random Forest, XGBoost, and MLP neural network models, the hybrid system becomes very powerful and surpasses current state-of-the-art. The solutions obtained an impressive accuracy of 97.57%, and this is more than the reinforcement learning (+7.9%), simple supervised (+4.1%), deep learning (+3.3%) and classical ensemble (+2.6%) methods. The model is based on probabilities and its soft voting mechanism is the key to its power, to keep the prediction confidence high while deflating the volatility. The model is integrated with key functional engineering - such as logarithmic returns, rolling volatility and volume-based indicators - and this synthesis displays complex non-linear market behavior even more challenging to model with traditional econometric models like GARCH. The chronological validation strategy adopted guarantees the assessment of the existing performance in the real world and the computational efficiency of the model allows its use in low latency, high-frequency trading environments. This study offers much scope for future research, which can further enhance the framework by extending it to multi-asset portfolio forecasts, and using non-traditional data sources such as news sentiment, measures of economic policy uncertainty and geopolitical risk indicators. The usage of advanced temporal architectures like Transformer, XLSTM etc. helps for better modeling of long term dependencies and using regime-switching techniques improves the robustness of the model when applied across various market conditions. Additionally, the combination of interpretability (e.g. SHAP, LIME) will not only make the model less of a black box, but also pave the way to create risk-adjusted trading strategies that will enable its using. Furthermore, future works will involve verification of trades in real life, development of ultra-fast optimization computationally, etc.

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